

1 PSPO methodology

Problem setting and overview. Provably Safe Policy Optimisation (PSPO) considers a finite (or finitely discretised) state space $\mathcal{S}$, a discrete action space $\mathcal{A}$, and a precomputed shield $H : \mathcal{S} \times \mathcal{A} \rightarrow \{0, 1\}$. The admissible actions are $\mathcal{A}_{\text{safe}}(s) = \{a : H(s, a) = 1\}$. PSPO trains a categorical policy π_θ with logits $z_\theta(s)$ while maintaining two complementary safety mechanisms. First, the shield makes data collection safe by replacing unsafe sampled actions before they reach the environment. Second, policy updates are constrained to certified *Local Invariant Domains* (LIDs): parameter orthotopes in which every policy’s greedy action is shield-admissible on every enumerated state. Thus the shield protects training-time interaction, whereas LIDs make the learned greedy policy safe without a shield. Algorithm 1 gives the default, region-first procedure; full objectives, default values, and proofs are in Appendix A.

Algorithm 1 PSPO at default settings (one enforcement interval)

Require: shield H ; environment; PPO actor π_θ ; exhaustive shield data $\mathcal{D}_{\text{sh}}$

- 1: $\theta_0 \leftarrow \text{SAFEINITIALISE}(\mathcal{D}_{\text{sh}})$; verify θ_0 exactly
- 2: **for** $k = 0, 1, \dots, K - 1$ **do**
- 3: $B_k \leftarrow \text{SHIELDEDROLLOUT}(\pi_{\theta_k}, H)$ $\triangleright$ execute safe; store proposed action
- 4: $\bar{\theta}_{k+1} \leftarrow \text{PPOUPDATE}(\theta_k, B_k)$
- 5: $\Delta_k \leftarrow \bar{\theta}_{k+1} - \theta_k$
- 6: $\beta_k \leftarrow \text{CALIBRATE SURROGATE}(\theta_k, \mathcal{D}_{\text{sh}})$
- 7: $\Theta_k \leftarrow \text{GROWDIRECTIONALLID}(\theta_k, \Delta_k, \beta_k, \mathcal{D}_{\text{sh}})$
- 8: **if** Θ_k has an exhaustive 100% IBP certificate **then**
- 9: $\theta_{k+1} \leftarrow \text{Proj}_{\Theta_k}^{\ell_2^2}(\bar{\theta}_{k+1})$
- 10: **else**
- 11: $\theta_{k+1} \leftarrow \theta_k$ $\triangleright$ fail closed
- 12: **return** greedy actor $\mu_{\theta_K}(s) = \arg \max_a z_{\theta_K, a}(s)$

1.1 Policy initialisation and surrogate calibration

The shield is converted to the exhaustive multi-label data set $\mathcal{D}_{\text{sh}} = \{(x(s), \mathbf{1}_{\mathcal{A}_{\text{safe}}(s)}) : \mathcal{A}_{\text{safe}}(s) \neq \emptyset\}$, where $x(s)$ is one-hot by default. A two-hidden-layer 64–64 Tanh actor is behaviourally cloned without reward. The loss (i) maximises total probability on the safe set, (ii) imposes the strict margin $\min_{a \in \mathcal{A}_{\text{safe}}(s)} z_a - \max_{b \notin \mathcal{A}_{\text{safe}}(s)} z_b \geq 2$, and (iii) encourages a uniform conditional distribution over safe actions. Training stops only after greedy safety is 100%, the margin target is met at every contested state, and the minimum normalised safe-action entropy is at least 0.95. This supplies a safe, non-degenerate centre θ_0 ; Appendix A.3 gives the loss and stopping rule.

LID growth uses a smooth LogSumExp (LSE) safety constraint. Before *each* LID computation, PSPO selects the smallest integer inverse temperature $\beta \in \{1, \dots, 1000\}$ (temperature $\tau = 1/\beta$) for which the point policy has a strictly positive LSE margin at every state. Recalibration is necessary because the scale of the logits changes during PPO. The calibrated surrogate is used to optimise the box, but acceptance always uses the hard IBP certificate

described below; see Appendix A.4.

1.2 Shielded exploration

At state s , PPO samples a proposal $\tilde{a} \sim \pi_\theta(\cdot | s)$. If $H(s, \tilde{a}) = 1$ it is executed unchanged; otherwise the shield samples uniformly from $\mathcal{A}_{\text{safe}}(s)$ and executes the replacement. By default, $\tilde{a}$ and its original log-probability are stored in the rollout buffer, while the reward and transition arise from the executed action. Shielding is confined to data collection: evaluation is unshielded and greedy, so its safety follows from the LID rather than hidden runtime intervention. Diagnostics record proposals, overrides, and executed-action violations (Appendix A.6).

1.3 Policy updates with safe LIDs

After a PPO phase proposes $\bar{\theta}_{k+1}$, PSPO grows an orthotope $\Theta_k = [\ell_k, u_k]$ about the last certified parameters θ_k . Growth is directional: if $\Delta_{k,j} > 0$ only $u_{k,j}$ may move, if $\Delta_{k,j} < 0$ only $\ell_{k,j}$ may move, and a zero coordinate remains fixed. Interval widths are weighted by $|\Delta_{k,j}|$, focusing the 200-step primal–dual growth budget on the proposed update. IBP bounds the logits jointly for every $\theta \in \Theta_k$; a box is retained only if, for every state,

$$\min_{a \in \mathcal{A}_{\text{safe}}(s)} z_a(s; \Theta_k) > \max_{b \notin \mathcal{A}_{\text{safe}}(s)} z_b(s; \Theta_k). \quad (1)$$

Optimisation stops early if the proposal is contained and this hard certificate holds. Otherwise the last fully certified checkpoint is used and the candidate is projected in ℓ_2 ; failed growth reverts to θ_k . The new LID replaces the old one. The default enforcement interval is one PPO rollout (100 for MiniPacman); consequently, the unshielded-policy guarantee holds at every enforced iterate. Details are in Appendix A.5.

1.4 Safety guarantees

Assume that the shield is sound and non-blocking, the certification data enumerate every state on which safety is claimed, IBP is sound, and deployment is greedy. Then Eq. (1) implies that every $\theta \in \Theta_k$ selects only a shield-safe action. Since projection (or fail-closed reversion) places every enforced iterate in a certified LID, induction gives $H(s, \mu_{\theta_k}(s)) = 1$ for all certified states and all enforced k . Independently, shielded exploration executes only admissible actions. If the shield is an inductive almost-sure safety shield and the initial state lies in its winning set, this yields trajectory-level almost-sure safety. The result is an *a priori* guarantee over the enumerated shield domain; it does not cover unshielded stochastic sampling, omitted states, or an unsound shield. Formal statements and boundary cases appear in Appendix A.8.

A Detailed PSPO Methodology

This appendix specifies the default PSPO configuration represented by Algorithm 1. It separates the safety statement from engineering details and makes explicit the assumptions under which the certificates compose with PPO.

A.1 Notation and safety specification

Let $\mathcal{S}_{\text{cert}} \subseteq \mathcal{S}$ be the finite set of states on which safety is certified. In the default tabular experiments it contains every state with at least one shield-admissible action. The shield is the binary relation

$$H(s, a) = 1 \iff a \in \mathcal{A}_{\text{safe}}(s). \quad (2)$$

States with $\mathcal{A}_{\text{safe}}(s) = \mathcal{A}$ impose no constraint. States with $\mathcal{A}_{\text{safe}}(s) = \emptyset$ are excluded from LID construction; a safety claim therefore requires the non-blocking assumption $\mathcal{A}_{\text{safe}}(s) \neq \emptyset$ for every state that can be visited. The categorical actor has logits $z_{\theta}(s) \in \mathbb{R}^{|\mathcal{A}|}$ and deterministic deployment policy

$$\mu_{\theta}(s) = \arg \max_{a \in \mathcal{A}} z_{\theta, a}(s). \quad (3)$$

PSPO uses the stronger default invariant

$$\phi_{\text{all}}(s, \theta) \text{equiv} \min_{a \in \mathcal{A}_{\text{safe}}(s)} z_{\theta, a}(s) > \max_{b \notin \mathcal{A}_{\text{safe}}(s)} z_{\theta, b}(s), \quad (4)$$

which makes *every* admissible logit dominate every inadmissible logit. This is stronger than merely requiring one admissible argmax; it also leaves PPO freedom to move probability among safe actions.

A.2 Default settings

Table 1 records the operational defaults. Environment-specific training horizons do not affect the safety argument. The default enforcement interval is one rollout for all reported tasks except MiniPacman, where it is 100 rollouts.

Table 1: PSPO operational defaults. “Exhaustive” means all shield states with at least one admissible action.

Component	Default
State representation	One-hot discrete state
Actor/critic	Two hidden layers, 64 units each, Tanh
Base-policy objective	Safe-set cross-entropy + all-safe margin + safe-action entropy
Initialisation targets	Margin $m = 2$; entropy weight 1; minimum normalised safe entropy 0.95
Base-policy optimiser	Adam; learning rate 3×10^{-3} ; batch size 512; at most 8000 epochs
PPO	Learning rate 3×10^{-4} ; rollout 2048; minibatch 64; 10 epochs; $\gamma = .99$; GAE $\lambda = .95$; clip .2; entropy coefficient 0; value coefficient .5; gradient norm .5
Exploration storage	Store proposed action and its log-probability; execute shielded action
Update rule	Region-first; replace previous LID; ℓ_2 projection; fail-closed reversion
LID geometry/direction	Orthotope; directional; $ \Delta $ -weighted widths; stop on certified containment
LID surrogate	All-safe-vs-all-unsafe LSE; inverse temperature recalibrated per computation
LID optimisation	IBP growth and certification; 200 iterations; checkpoint every 100; full-data batch
Certification	Exhaustive states; require minimum hard certified accuracy 1.0
Enforcement	Every PPO rollout by default; every 100 rollouts for MiniPacman

A.3 Safe policy initialisation

A.3.1 Shield demonstration data

For every $s \in \mathcal{S}_{\text{cert}}$, let $x(s)$ be its policy input and let $y_{s,a} = \mathbf{1}[a \in \mathcal{A}_{\text{safe}}(s)]$. The reward-free data are

$$\mathcal{D}_{\text{sh}} = \{(x(s), y_s) : s \in \mathcal{S}_{\text{cert}}\}. \quad (5)$$

The default uses one-hot $x(s)$; decoded features may be used if the mapping between features and shield-state indices is supplied. The actor and the PPO policy must have the same architecture so that the fitted parameters can be copied exactly.

A.3.2 Behaviour-cloning objective

Write $P_\theta(\mathcal{A}_{\text{safe}}(s) \mid s) = \sum_{a \in \mathcal{A}_{\text{safe}}(s)} \pi_\theta(a \mid s)$. The safe-set cross-entropy is

$$\mathcal{L}_{\text{set}}(\theta) = -\frac{1}{|\mathcal{D}_{\text{sh}}|} \sum_s \log P_\theta(\mathcal{A}_{\text{safe}}(s) \mid s) = \frac{1}{|\mathcal{D}_{\text{sh}}|} \sum_s [\text{LSE}_a z_{s,a} - \text{LSE}_{a \in \mathcal{A}_{\text{safe}}(s)} z_{s,a}]. \quad (6)$$

For a contested state (one having at least one unsafe action), define the all-safe margin

$$d_s(\theta) = \min_{a \in \mathcal{A}_{\text{safe}}(s)} z_{s,a} - \max_{b \notin \mathcal{A}_{\text{safe}}(s)} z_{s,b}, \quad (7)$$

and the hinge $\mathcal{L}_{\text{mar}} = |C|^{-1} \sum_{s \in C} [m - d_s]_+$ with $m = 2$. Finally, condition the policy on its safe actions,

$$q_{s,a} = \frac{\pi_\theta(a \mid s)}{P_\theta(\mathcal{A}_{\text{safe}}(s) \mid s)}, \quad a \in \mathcal{A}_{\text{safe}}(s), \quad (8)$$

and define normalised entropy

$$\bar{H}_s(q) = \frac{-\sum_{a \in \mathcal{A}_{\text{safe}}(s)} q_{s,a} \log q_{s,a}}{\log |\mathcal{A}_{\text{safe}}(s)|}. \quad (9)$$

Single-safe-action states are assigned $\bar{H}_s = 1$ and omitted from the entropy loss. The default training loss is

$$\mathcal{L}_{\text{init}} = \mathcal{L}_{\text{set}} + \mathcal{L}_{\text{mar}} + \frac{1}{|C_{>1}|} \sum_{s \in C_{>1}} (1 - \bar{H}_s), \quad (10)$$

with unit weights. Training stops only when: (i) the greedy action is safe on every row of $\mathcal{D}_{\text{sh}}$; (ii) $\min_{s \in C} d_s \geq 2$; and (iii) $\min_{s \in C_{>1}} \bar{H}_s \geq 0.95$. Failure to meet all three conditions within the budget aborts the PSPO pipeline. This is stricter than plain behavioural cloning and avoids a safe but nearly tied initial policy.

A.4 Per-LID surrogate calibration

The constrained optimiser needs a differentiable safety margin. For temperature $\tau > 0$, define

$$\text{smin}_\tau(v_A) = -\tau \log \sum_{a \in A} \exp(-v_a/\tau), \quad (11)$$

$$\text{smax}_\tau(v_B) = \tau \log \sum_{b \in B} \exp(v_b/\tau). \quad (12)$$

At a point policy, the default all-safe LSE surrogate is

$$g_s(\theta; \tau) = \text{smin}_\tau(z_{\theta, \mathcal{A}_{\text{safe}}(s)}(s)) - \text{smax}_\tau(z_{\theta, A \setminus \mathcal{A}_{\text{safe}}(s)}(s)). \quad (13)$$

Because $\text{smin}_\tau(v) \leq \min v$ and $\text{smax}_\tau(v) \geq \max v$, $g_s > 0$ is sufficient for Eq. (4). Immediately before each LID computation PSPO evaluates the last certified point θ_k and chooses

$$\beta_k = \min \left\{ \beta \in \{1, \dots, 1000\} : \min_s g_s(\theta_k; 1/\beta) > 0 \right\}. \quad (14)$$

If this set is empty, calibration fails and the proposed update is rejected. Calibration is repeated because multiplying all logits changes the numerical sharpness needed by the surrogate even though it leaves the greedy action unchanged.

A.5 Directional safe-LID construction

A.5.1 Parameterisation and direction masks

Let the last certified parameters be θ_k and let the PPO candidate be $\bar{\theta}_{k+1}$. A LID is an orthotope

$$\Theta_k = [\ell_k, u_k] = \{\theta : \ell_{k,j} \leq \theta_j \leq u_{k,j} \ \forall j\}, \quad \ell_k \leq \theta_k \leq u_k. \quad (15)$$

Writing $\Delta_k = \bar{\theta}_{k+1} - \theta_k$, directional growth fixes

$$\begin{cases} \ell_{k,j} = \theta_{k,j}, & \Delta_{k,j} > 0, \\ u_{k,j} = \theta_{k,j}, & \Delta_{k,j} < 0, \\ \ell_{k,j} = u_{k,j} = \theta_{k,j}, & \Delta_{k,j} = 0. \end{cases} \quad (16)$$

Thus the box grows only into the orthant containing the candidate.

A.5.2 IBP safety constraint and growth objective

Interval bound propagation maps the parameter box and the point input $x(s)$ to sound logit intervals

$$z_{\theta,a}(s) \in [\underline{z}_a(s; \Theta_k), \bar{z}_a(s; \Theta_k)] \quad \text{for all } \theta \in \Theta_k. \quad (17)$$

The lower LSE margin used during growth is

$$\underline{g}_s(\Theta_k; \tau_k) = -\tau_k \log \sum_{a \in \mathcal{A}_{\text{safe}}(s)} \exp\left(-\frac{\underline{z}_a(s; \Theta_k)}{\tau_k}\right) - \tau_k \log \sum_{b \notin \mathcal{A}_{\text{safe}}(s)} \exp\left(\frac{\bar{z}_b(s; \Theta_k)}{\tau_k}\right). \quad (18)$$

The default objective weights each scalar interval by the magnitude of the candidate update. With $w_j = |\Delta_{k,j}|$ and width $r_j = u_{k,j} - \ell_{k,j}$, the current default ($\alpha = 0$) is

$$J(\ell_k, u_k; \Delta_k) = \frac{\sum_j w_j r_j}{\sum_j w_j}, \quad (19)$$

where a zero denominator gives the differentiable objective zero. The implementation supports a log-volume mixture $\alpha \sum_j w_j \log(r_j + 10^{-6}) + (1 - \alpha) \sum_j w_j r_j$, but the default is pure weighted width. Conceptually, LID growth solves

$$\max_{\ell_k, u_k} J(\ell_k, u_k; \Delta_k) \quad \text{s.t.} \quad \underline{g}_s(\Theta_k; \tau_k) \geq 0 \quad \forall s \in \mathcal{S}_{\text{cert}}, \quad (20)$$

subject to Eq. (16), valid lower/upper ordering, and a maximum scalar width of 10 on either side of the centre. The implementation uses alternating Adam primal and SGD dual steps in an augmented-Lagrangian formulation. It runs at most 200 iterations and records checkpoints every 100 iterations.

A.5.3 Hard certification and checkpoint selection

The soft constraint guides optimisation but is not the returned certificate. Each saved box is rebuilt as an IBP model and receives the hard per-state test

$$c_s(\Theta_k) = \mathbf{1} \left[\min_{a \in \mathcal{A}_{\text{safe}}(s)} \underline{z}_a(s; \Theta_k) > \max_{b \notin \mathcal{A}_{\text{safe}}(s)} \bar{z}_b(s; \Theta_k) \right]. \quad (21)$$

All states are used both as the optimisation batch and the certification batch. A checkpoint qualifies only when $\min_s c_s = 1$; PSPO selects the last qualifying checkpoint. If the growing box contains $\bar{\theta}_{k+1}$ and passes this same exhaustive hard test, growth stops early. If no checkpoint qualifies, PSPO returns no LID and reverts to θ_k .

A.6 Shielded exploration and PPO semantics

Let $\tilde{a}_t$ be the categorical action sampled by PPO. The executed action is

$$a_t = \begin{cases} \tilde{a}_t, & H(s_t, \tilde{a}_t) = 1, \\ \text{Uniform}(\mathcal{A}_{\text{safe}}(s_t)), & H(s_t, \tilde{a}_t) = 0. \end{cases} \quad (22)$$

The default rollout tuple stores $(s_t, \tilde{a}_t, r_t, \log \pi_{\theta}(\tilde{a}_t | s_t))$, but r_t and s_{t+1} result from a_t . This preserves a direct learning signal about the policy’s proposal and matches the implemented default, although it means the collected transition is generated under the shielded behaviour policy. An optional non-default mode stores a_t and recomputes its log-probability. In either mode, only rollout collection is shielded: the ordinary prediction method remains unshielded.

A.7 Projection and enforcement frequency

For a certified orthotope, Euclidean projection is coordinate-wise clipping,

$$[\text{Proj}_{\Theta_k}^{\ell_2}(\bar{\theta})]_j = \min\{u_{k,j}, \max\{\ell_{k,j}, \bar{\theta}_j\}\}. \quad (23)$$

After projection, the result becomes the next last-safe snapshot. In region-first mode the previous LID is replaced, rather than unioned, so each guarantee relies only on the newest certificate. A failed calibration, invalid candidate, failed LID computation, or exhausted budget restores the last-safe parameters.

With enforcement interval $q = 1$, PPO performs its minibatch epochs, then PSPO certifies and projects before the next rollout. If $q > 1$, shielded rollouts remain action-safe, but the raw actor between enforcement boundaries need not lie in a LID. Thus the unshielded actor should be exported or evaluated only after calling the final enforcement step. MiniPacman uses $q = 100$; the other default environments use $q = 1$.

A.8 Formal guarantees and scope

Assumptions. The following conditions make the guarantee explicit.

- (A1) $\mathcal{S}_{\text{cert}}$ exhausts the states on which safety is claimed, and every such state has at least one admissible action.
- (A2) The shield is sound: $H(s, a) = 1$ implies that a satisfies the intended one-step safety specification at s .
- (A3) The IBP implementation returns sound logit bounds for the stated network, input, and parameter orthotope.
- (A4) Unshielded deployment is deterministic and greedy; safety enforcement is completed before evaluation or export.
- (A5) Projection is applied to exactly the actor parameters certified by IBP, with consistent parameter ordering and numerically valid interval bounds.

Lemma 1 (LSE soundness). If $\underline{g}_s(\Theta; \tau) > 0$ in Eq. (18), then every $\theta \in \Theta$ satisfies Eq. (4) at s .

Proof. For any finite vector v , $-\tau \log \sum_i \exp(-v_i/\tau) \leq \min_i v_i$ and $\tau \log \sum_i \exp(v_i/\tau) \geq \max_i v_i$. Sound IBP additionally gives $z_{\theta,a} \geq \underline{z}_a$ for safe actions and $z_{\theta,b} \leq \bar{z}_b$ for unsafe actions. Therefore a positive lower LSE margin implies $\min_{a \in \mathcal{A}_{\text{safe}}(s)} z_{\theta,a} > \max_{b \notin \mathcal{A}_{\text{safe}}(s)} z_{\theta,b}$ for every $\theta \in \Theta$. $\square$

Theorem 1 (safe enforced PSPO iterates). Under Assumptions (A1)–(A5), suppose θ_0 satisfies Eq. (4) on $\mathcal{S}_{\text{cert}}$. If every PSPO enforcement either (i) projects the PPO proposal into an orthotope satisfying Eq. (21) for all $s \in \mathcal{S}_{\text{cert}}$, or (ii) reverts to the previous certified parameters, then for every enforced iterate k ,

$$H(s, \mu_{\theta_k}(s)) = 1 \quad \forall s \in \mathcal{S}_{\text{cert}}. \quad (24)$$

Proof. The claim holds for $k = 0$ by verified initialisation. If enforcement reverts, the inductive hypothesis is unchanged. Otherwise Eq. (21), IBP soundness, and membership of the projection in the certified orthotope imply strict safe-over-unsafe logit ordering at every certified state. Hence the greedy argmax is admissible. Induction over enforcement steps proves Eq. (24). $\square$

Corollary 1 (safe exploration actions). Under Assumptions (A1) and (A2), every action executed during shielded exploration is admissible, regardless of whether the current proposal policy lies in a LID. This follows immediately from Eq. (22).

Corollary 2 (trajectory safety under an inductive shield). If, in addition, the shield’s winning set is forward invariant under every admissible action and the initial state belongs to that set, then shielded exploration is almost-surely safe for the full trajectory. The same statement applies to unshielded greedy execution at enforced PSPO iterates because Theorem 1 makes those actions shield-admissible.

Scope and exclusions. The guarantee is deterministic and domain-specific. It does not assert safety for states omitted from $\mathcal{S}_{\text{cert}}$, for an incorrect shield, or for unshielded stochastic sampling (which can select a low-probability unsafe action despite a safe argmax). When enforcement is less frequent than one rollout, the raw policy is certified only at enforcement boundaries; shielded interaction remains protected in between. Finally, the theorem concerns safety, not reward improvement: projection can reduce or entirely reject a PPO update.